What a Single Accuracy Number Hides: Separating Value-Level Wandering from Execution-Level Failure in Physics Reasoning

Anonymous Authors

Abstract

A single accuracy number cannot tell a model that computes from a model that recognises the problem family — and, we show, it also cannot tell a stable model from one whose answers wander. We audit what the headline metric misses with five probes plus two controls on physics reasoning: a value-level perturbation (rescale the numbers), a correctness-level perturbation (rephrase the question while leaving physics and data bit-identical), a multiple-choice format control, corrupt-and-restore causal tracing, and an error-type annotation checked by a second, cross-family judge. Across 323 SciBench physics problems and five open-weight models, three results do the work. First, on the measurement side, paraphrase changes 30.4% of final *values* yet moves only 5 of 240 items across the correct/incorrect boundary, mostly in the *favourable* direction (4 items are rescued by the paraphrase versus 1 broken; exact McNemar $p = 0.375$): retrospective re-scoring of the same generations converts what a value-level comparison reports as a large difficulty effect into a non-significant one with no systematic penalty. Second, on the mechanism side, output execution and semantic correctness run on different pathways: parseability is essentially layer-invariant while correctness is depth-dependent, and on problems the model cannot solve, restoring any single layer never exceeds 3.3% (Wilson 95% CI $[1.1, 9.3]$). Third, recoverable behaviour occupies a *contiguous early-middle band* (layers 4–14 of 28), and we show that single-layer claims are *not identifiable* at our sample size — a negative identifiability result we treat as a finding, not an apology. Two validity bounds accompany these: strict-format evaluation and fallback evaluation give different ceilings on the same generations (71%–90% of outputs never emit the requested answer marker), and two LLM judges agree at only $\kappa = 0.133$ on an error taxonomy whose E1/E2 boundary is not textually observable. Everything runs on one 80 GB GPU in about 2.6 GPU-hours, and the dual-annotated sample is released.

1 Introduction

Chain-of-thought prompting [13] is routinely credited with giving language models "reasoning" ability, and physics benchmarks such as SciBench [12] and OlympiadBench [4] are used to demonstrate progress. The evidence offered is almost always the same shape: a single accuracy number, compared across models or prompts. That number silently aggregates decisions about what counts as the answer, which generations count as valid, and which differences are real. This paper takes those decisions apart.

Thesis: the metric is the object of study. This is not a "which model is better at physics" paper. It is an audit of what the headline metric — accuracy on a benchmark, compared across prompts, formats and models — can and cannot support. Three things a single accuracy number cannot distinguish, and that we measure separately: (i) an answer *value* that drifts under a semantics-preserving paraphrase from an answer whose *correctness* changes; (ii) producing *a* number from producing *the right* number; and (iii) a behaviour that is *localisable* to a layer range from one that would licence a claim about a specific layer. In each case we find that the metric as usually reported hides a distinction that changes the interpretation, and in two of the three the honest conclusion is a *validity bound* on the evaluation rather than a statement about the model.

Our probes are deliberately cheap and falsifiable. A perturbation that changes the physics (rescale the numbers) should move a computing model; a perturbation that changes only the phrasing should not.

Causal tracing should localise whatever computation exists. An error taxonomy should agree with itself across annotators. Where a probe returns a negative or uninformative answer, we report the bound it imposes rather than the conclusion we would have liked.

- **Baseline behaviour.** (§4.1) Five open-weight models on 323 SciBench physics problems, with bootstrap intervals and the prompt/scale interaction made explicit.
- **Value vs. correctness.** (§4.2) A paraphrase that leaves physics and data bit-identical, re-scored against the reference rather than compared answer-to-answer.
- **Execution vs. semantics.** (§4.3) Corrupt-and-restore tracing, with two metrics recorded per layer: is the emitted answer parseable, and is it correct?
- **Knowledge vs. format.** (§4.4) A multiple-choice control that holds the physics fixed and changes only the output format.
- **Validity bounds.** Error attribution across two judges (§4.5), prompt scaffolding (§4.6), and an output-format ceiling that caps what strict grading can see (§4.7).

Contributions.

1. **A measurement claim, not a model claim** (strongest result): on the same generations, a semantics-preserving paraphrase changes 30.4% of answer *values* but moves only 5 of 240 items across the correct/incorrect boundary, mostly in the favourable direction and not significant under an exact McNemar test. Any evaluation that compares prompt variants by a single accuracy pass — or by answer agreement — can therefore read value-level wandering as a difficulty change. We show the two readings diverge, and that the direction of the error is predictable: agreement-based comparisons overstate the effect.
2. **An identifiability result about layer localisation**: recoverable behaviour forms a contiguous early-middle band (layers 4–14 of 28), and single-layer claims are *not identifiable* at our sample size — the point-estimate maximum is statistically indistinguishable from several other layers inside the band. We report the band, the formal definition that produces it, and the identifiability limit as a result rather than as a caveat.
3. **A representation-vs-output decomposition**: parseability is essentially layer-invariant (96% even under corruption), while correctness is depth-dependent; and on problems the model cannot solve, restoration never rescues them at any single layer ($\leq$3.3%). Execution and semantics are separable, and the failures are capability limits rather than misplaced representations.
4. **Two validity bounds on the measurement apparatus**, reported alongside (not instead of) the findings: between 71% and 90% of outputs never emit the requested answer marker, so strict-format and fallback grading give different ceilings on identical generations; and two LLM judges agree at only $\kappa = 0.133$ on an error taxonomy whose E1/E2 boundary is not textually observable. We release the dual-annotated sample as a resource for LLM-as-a-judge reliability research.

2 Related Work

Physics and science reasoning benchmarks. SciBench [12] collects college-level textbook problems with free-response answers; we use its physics subsets and restrict to the 323 problems whose reference answers are numerically gradable. OlympiadBench [4] contains competition problems; only 43% of its physics items have numerically gradable references, a coverage limit we carry through every table. MMLU [5] provides multiple-choice knowledge questions and serves here as a format control rather than as a target.

Causal tracing and localisation. Activation- and state-level interventions are standard for localising behaviour in open models [8, 11]. We use the simplest variant — corrupt the input, then restore the clean residual stream at a single layer — because it requires no training and yields a per-layer curve. In factual-recall studies the recoverable signal typically sits in the middle of the network — for a 28-layer model, roughly the central third (layers ∼9–19); our plateau sits *earlier* than that, which we discuss in §4.3.

Sensitivity and robustness of LLM answers. Prompt sensitivity is well documented [10], usually as a nuisance to be averaged out. We instead treat sensitivity as the measurement, and — unlike most prior work — we separate *value* changes from *correctness* changes, which turn out to have very different magnitudes.

LLM-as-a-judge and error taxonomies. Using a strong model to label errors is common practice [14], and its reliability is usually reported as agreement with human labels. We contribute a case where two *LLM* judges disagree substantially on a subjective taxonomy, and argue that the disagreement is itself the informative quantity.

3 Method

3.1 Benchmarks and grading

All correctness decisions are made by symbolic/numeric comparison. The final answer is extracted by a precedence-ordered procedure (`ANSWER:` line, taking the last occurrence; then `\boxed{}`; then "answer is X"; then the last number on the last line), normalised for Unicode minus signs, superscripts and thousands separators, and compared to the reference under a 5% relative tolerance (0.01 absolute for near-zero targets).

Reference answers are parsed with a *stricter* parser than model outputs: a reference that is not a single number is marked unscorable and removed from the denominator rather than counted as a model error, because a permissive parser reads expressions such as $x^2 + 2x$ as 2 and silently corrupts the denominator. For SciBench physics all 323 references are numerically gradable, so the denominator is the full set; for OlympiadBench only 152 of 351 are. Grading is covered by 23 self-test cases and 84 unit tests, and an oracle check (feed the reference as the model output) yields 100% self-consistency. Full prompt templates are given in Appendix B.

3.2 Models and inference

We evaluate five open-weight instruct models: Qwen2.5-3B/7B/14B-Instruct [9], Llama-3.1-8B-Instruct [2], and DeepSeek-R1-Distill-Qwen-7B [3]. Decoding is greedy with a fixed seed, so every reported number is deterministic. We use the tokenizer's chat template and left padding for batched generation. All runs use the `transformers` backend: vLLM was *not installed* in our execution environment, so the code path that prefers vLLM never activated and no result mixes backends. Under greedy decoding accuracy is backend-independent; only wall-clock differs (§6).

3.3 Perturbation probes

Each probe is a deterministic transformation of the problem text:

- **numeric**: scale physical values (physics intact, at most two numbers changed, exponents and power bases protected);
- **wording**: paraphrase the framing only, leaving physics and data bit-identical. This is the only variant whose reference answer remains valid, so it is the only one for which we report a paired correctness comparison;
- **swap**: exchange two distinct physical quantities, producing a physically inconsistent problem.

Because the correct answer to a *numeric* or *swap* variant is not directly computable, those variants are scored by the distribution of output change, and we report median and p90 rather than the mean (the sensitivity score $|\Delta|/(|base| + \epsilon)$ is dominated by outliers when the base answer is near zero). The *swap* probe covers only 36 of 323 problems, so its percentages are reported for completeness and carry no quantitative weight.

3.4 Causal tracing

The numeric quantities of a question are replaced by a semantically empty placeholder; the corrupted prompt is run; then the clean residual stream is *restored* at exactly one layer at a time, all other layers remaining corrupted. Restoration is applied during prefill only and never to decode steps, and clean and corrupted

Table 1: Accuracy (%) on SciBench physics (323 problems), and what the accuracy matrix hides. All reference answers are numerically gradable and all model outputs were parseable, so the denominator is the full 323. Bracketed intervals are bootstrap 95% CIs (10,000 resamples, seed=42) on overall accuracy; sub-domain cells are point estimates. The last two columns expose the prompt/scale interaction that the raw accuracies do not: Δ is the cot−direct difference in points and the ratio is cot/direct. Note that the smallest Qwen model still beats the larger Llama model, and that Llama gains far less from chain-of-thought.

Model	Mode	Overall (95% CI)	Sub-domain accuracy (%)						cot vs direct	
			Math	Mech	Thermo	Optics	Quantum	General	Δ (pp)	ratio
Qwen2.5-3B	cot	12.1 [8.7, 15.8]	20.0	5.4	12.1	10.6	18.2	9.9	+6.8	2.28×
Qwen2.5-3B	direct	5.3 [3.1, 7.7]	2.0	7.1	7.6	6.4	6.1	2.8		
Qwen2.5-7B	cot	21.1 [16.7, 25.7]	34.0	21.4	21.2	17.0	15.2	16.9	+12.4	2.43×
Qwen2.5-7B	direct	8.7 [5.6, 11.8]	8.0	10.7	6.1	8.5	12.1	8.5		
Qwen2.5-14B	cot	25.7 [21.1, 30.7]	40.0	26.8	24.2	23.4	21.2	19.7	+13.9	2.18×
Qwen2.5-14B	direct	11.8 [8.4, 15.5]	6.0	17.9	7.6	10.6	30.3	7.0		
Llama-3.1-8B	cot	9.9 [6.8, 13.3]	14.0	8.9	12.1	6.4	6.1	9.9	+2.2	1.29×
Llama-3.1-8B	direct	7.7 [5.0, 10.8]	6.0	7.1	6.1	8.5	18.2	5.6		
R1-Distill-7B	cot	11.1 [7.7, 14.6]	12.0	7.1	10.6	8.5	18.2	12.7	–	–

sequences are aligned from the right because they tokenise to different lengths. Only problems whose clean run was correct *and* whose corrupted run lost score enter the recovery rate, which we report together with the skip counts.

4 Results

The order of the subsections follows the audit: each starts from a number that is normally reported and then reports what that number conceals. §4.1 establishes the baseline and shows the prompt×scale interaction the accuracy matrix hides; §4.2 separates value drift from correctness change; §4.3 separates output execution from semantic correctness and states an identifiability limit; §4.4 holds knowledge fixed while varying format; §4.5 and §4.7 are the two validity bounds; §4.6 closes the loop by testing a mitigation. Read in order, the sections support one claim: the standard single-number summary is under-specified for the distinctions this benchmark is used to draw.

4.1 How large is the prompting effect relative to scale?

Table 1 reports SciBench physics accuracy for all five models under both prompting conditions, with bootstrap 95% confidence intervals.

In this setting the prompting effect is large relative to a doubling of parameters, although the two are measured on different scales (a ratio versus a difference in points), so the comparison should not be read as a magnitude ranking: for the Qwen2.5 family chain-of-thought multiplies accuracy by 2.28×, 2.43× and 2.18× at 3B, 7B and 14B, while going from 7B to 14B adds 4.6 points (21.1% → 25.7%). The prompting benefit is family-dependent rather than a law of nature: for Llama-3.1-8B the same comparison gives only 1.29× (9.9% vs 7.7%), and family can outweigh size — the 3B Qwen model (12.1%) outperforms the 8B Llama model (9.9%). The reasoning-distilled model (DeepSeek-R1-Distill-Qwen-7B, 11.1%) does not beat its same-size instruct sibling, which we attribute to friction between long reasoning traces and this benchmark's terse numeric answer format; we did not run it answer-only, because its instruction-tuned behaviour emits reasoning traces regardless of the prompt and a *direct* comparison would not be like-for-like.

What the accuracy number hides here. The matrix in Table 1 reports accuracies; the interaction that matters is in its last two columns, which no single cell expresses. Chain-of-thought helps the Qwen family by 2.28–2.18× but Llama-3.1-8B by only 1.29×, so "prompting helps" is family-conditional, and a paragraph comparing two models at one prompt condition can reach the opposite conclusion depending on which condition is chosen. Likewise the family-over-scale inversion (Qwen 3B at 12.1% versus Llama 8B at 9.9%) is invisible if a study samples one model per family and reports a single score. Neither point is new in kind, but both are exactly what a headline number discards.

Table 2: Sensitivity of final answers to the two perturbations that carry weight (SciBench physics, 323 base problems, answer-only prompt). *numeric* rescales physical values with the physics intact; *wording* rewrites surface phrasing only, leaving physics and data bit-identical. "Changed" is the share of answers that differ from the base run — a value-level statistic, which is why Table 3 re-scores the same items against the reference. Median / p90 are reported instead of the mean: the sensitivity score is $|\Delta|/(|base| + \epsilon)$ and is dominated by outliers when $|base| \approx 0$.

Model	Variant	Coverage	n parsed	Median sens.	p90 sens.	Changed
Qwen2.5-3B	numeric	305/323 (94.4%)	305	0.500	6.32	77.0%
	wording	240/323 (74.3%)	240	0.000	0.91	33.8%
Qwen2.5-7B	numeric	305/323 (94.4%)	305	0.587	8.96	77.7%
	wording	240/323 (74.3%)	240	0.000	0.95	30.4%
Qwen2.5-14B	numeric	305/323 (94.4%)	305	0.633	7.60	83.9%
	wording	240/323 (74.3%)	240	0.000	0.68	31.7%
Llama-3.1-8B	numeric	305/323 (94.4%)	303	0.082	5.11	57.4%
	wording	240/323 (74.3%)	238	0.000	1.00	24.4%

A third probe (*swap*, exchanging two distinct physical quantities) covers only a small fraction of the benchmark and is reported in Appendix A; it is not used for any quantitative claim.

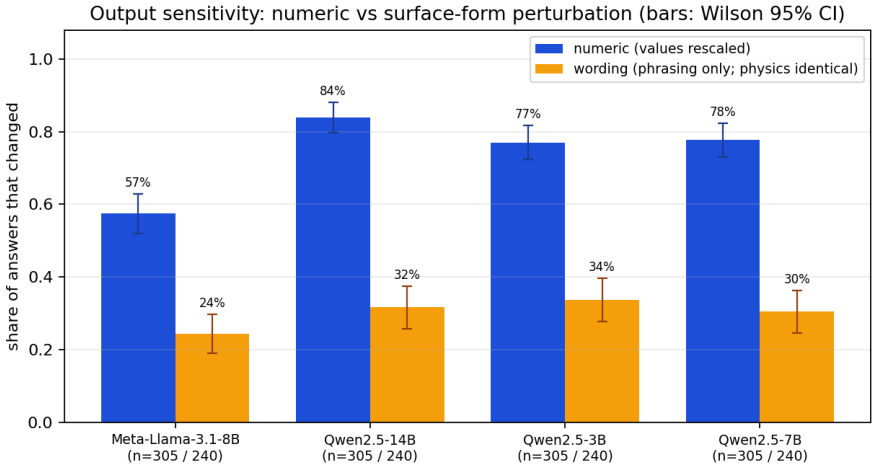


Figure 1: Share of final answers that change when physical values are rescaled (*numeric*, 305 items) versus when only the phrasing is rewritten (*wording*, 240 items), out of 323; bars show Wilson 95% intervals. The paraphrase bar is non-zero for every model — but §4.2 shows this is a value-level effect, not a correctness-level one.

4.2 What the accuracy number hides, I: value drift vs. correctness change

Table 2 reports the two perturbation probes that carry quantitative weight for four models (a third, *swap*, covers too few items to support any claim and is reported in Appendix A).

Changing the numbers moves the answer most of the time (77.7% for Qwen2.5-7B), so the stated values do participate in the computation: the model is not a lookup table. A paraphrase that leaves physics and data bit-identical changes 30.4% of answers, and the pattern repeats across models (33.8%, 31.7%, 24.4%); Figure 1 visualises the contrast.

Crucially, "changing the answer" is not the same as "changing correctness". Table 3 scores the paraphrase variants against the reference and reports the paired 2×2 counts. For Qwen2.5-7B the number of *discordant* items — those crossing the correct/incorrect boundary, which is by construction 1 + 4 — is 5 out of 240: 1 went correct → wrong and 4 went wrong → correct (exact McNemar $p = 0.375$). The pattern is the same in kind for the other three models, though the direction of the few discordant items varies. We therefore state the weaker and better-supported claim: we detect no systematic *penalty* from paraphrase — if anything the discordant items lean favourable — and the effect is not significant in either direction. The paraphrase instability is real, but it is *value-level wandering* rather than a systematic loss of competence — a distinction that a single accuracy number hides entirely.

Table 3: Paraphrase variants scored for *correctness*: 2×2 paired counts. The perturbation rewrites phrasing only, so the reference answer stays valid and a paired comparison is meaningful. Very few items cross the correct/incorrect boundary, in small numbers, and not consistently in one direction (see the two middle columns), i.e. we do not detect a systematic shift in either direction (exact McNemar test on the discordant pairs).

Model	both correct	correct→wrong	wrong→correct	n discordant	exact p
Qwen2.5-3B	8	5	5	10	1.000
Qwen2.5-7B	19	1	4	5	0.375
Qwen2.5-14B	27	1	6	7	0.125
Llama-3.1-8B	19	1	0	1	1.000

240 items per model (the paraphrase coverage). A value can change without changing correctness: the much larger 'changed answer' rate in Table 2 counts value changes, whereas this table counts only changes that cross the correctness boundary.

Table 4: Causal tracing on Qwen2.5-7B-Instruct (28 layers): recovery rate when the clean residual stream is restored at a single layer of the corrupted run. Only problems whose clean run was correct *and* whose corrupted run actually lost score are counted ($n = 14$ of 150 attempted; skipped 134 clean-wrong and 2 corrupted-flat), so the estimate applies to problems the model can already solve. Intervals are bootstrap 95% CIs over problems (10,000 resamples, seed 42).

Layer range	Mean recovery	Bootstrap 95% CI	±1 std across problems
layers 0–2	+0.431	[-0.11, +0.84]	±0.912
layers 4–8†	+0.665	[+0.33, +0.95]	±0.378
layers 10–16	+0.614	[+0.28, +0.88]	±0.431
layers 18–22	+0.238	[+0.00, +0.64]	±0.337
layers 24–26	+0.061	[+0.00, +0.19]	±0.174

†Recovery is maximal at layer 8 (+0.837, 95% CI [+0.69, +0.95]), but a paired bootstrap over problems *cannot* separate it from 6 other layers inside the plateau. We therefore claim a recoverable *range* (roughly layers 4–16), not a single peak layer; only layers beyond ∼18 and the embedding region (whose variance is an order of magnitude larger) are clearly distinguishable.

One caveat on that characterisation: we did not stratify the discordant items by the magnitude of the base answer. If they concentrate on near-zero answers — where the sensitivity score is unstable by construction (§3) — the "wandering" reading would need qualification, since such items are also the ones most likely to sit near a grading boundary. We flag this as untested rather than assume it away.

What the accuracy number hides here. A single accuracy pass over the base and paraphrased prompts yields 8.3% and 9.6% (Table 3), a difference that a careless comparison could report as an effect of prompt wording. Re-scoring the same generations against the reference shows the difference rests on 5 discordant items out of 240, is not significant, and reverses sign across models (0.0 / 1.2 / 2.1 / -0.4 points). Two distinct comparisons are available — *answer agreement* with the base run, and *correctness* against the reference — and they disagree by an order of magnitude. Reporting only the first is not a conservative error: it manufactures an effect.

4.3 Where is the computation, and is the output on the same pathway?

Table 4 locates the computation in depth.

Scope. Of 150 attempted problems, only 14 satisfy the strict inclusion rule (134 skipped as clean-wrong, 2 as corruption-flat). This subset is strongly selected: it consists of problems the model *already solves*. Every statement in this subsection is therefore scoped to solvable problems, not to the benchmark as a whole.

A band, not a peak — and that is the result. Recovery rises across the early-middle layers and collapses after layer 18 (Figure 2). The point estimate is maximal at layer 8 (+0.84), with a bootstrap 95% CI on that mean of [+0.69, +0.95]. A paired bootstrap over problems, however, cannot separate it from 6 other layers.

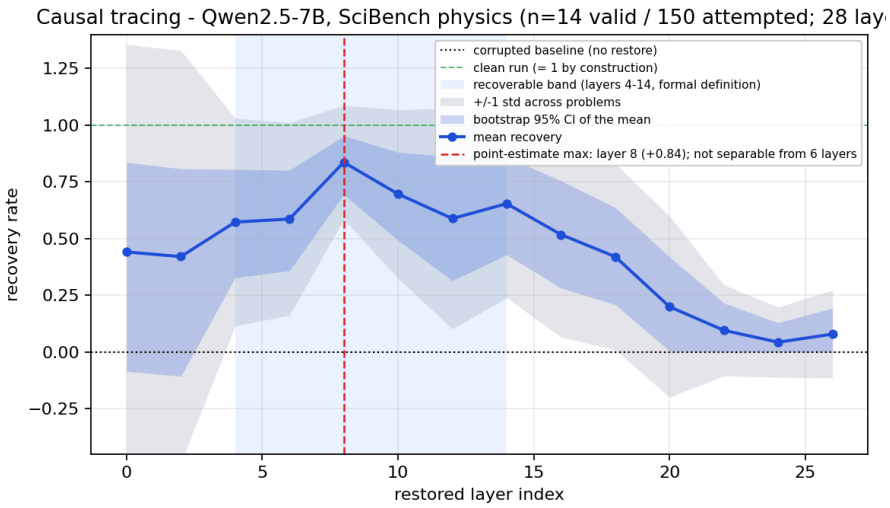


Figure 2: Recovery rate as a function of the restored layer, with ±1 standard deviation across problems. The dotted line at zero is the corrupted (no-restore) baseline, so the height above it is what restoration buys. Behaviour is recoverable over an early-middle band.

We regard this as an identifiability result rather than a failed localisation. At 14 exploitable problems, single-layer attribution is not supportable in this setting: the data licence a *range* claim and nothing finer. Studies that report a specific critical layer from comparable sample sizes are, on our estimate, reporting a point estimate whose uncertainty is larger than the differences between adjacent layers — which is a statement about what the protocol can establish, not about the model alone.

How the band is defined. Rather than reading a range off the curve by eye, we define it formally: the plateau is the maximal *contiguous* run of layers whose recovery is not separable from the peak under the paired bootstrap. This yields layers 4–14 (6 sampled layers); the next sampled layer, 16, *is* separable from the peak, which is why the band stops there. Because layer sampling is quantised (every 2 layers), the boundaries are accurate to one sampling step. We therefore claim a recoverable *band* of layers 4–14 rather than a single peak layer; only layers beyond ∼18, and the embedding region (whose variance is an order of magnitude larger), are clearly distinguishable.

Relation to factual-recall localisation. In factual-recall tracing the recoverable signal typically occupies the middle of the network (roughly the central third of depth). Our plateau (layers 4–14 of 28, i.e. the first half) sits *earlier* than that. Two differences plausibly explain the discrepancy. First, our corruption replaces *numerical quantities* with a placeholder, whereas factual-recall protocols replace a *named subject*; collapsing the numeric input removes information that has to be re-encoded early, so restoration must intervene early. Second, the target here is a derived scalar rather than a retrieved attribute, so the computation is spread over more layers. These are hypotheses consistent with our data, not findings we can establish without additional corruption types.

Separating execution from semantics. Restoring a clean state is only defined for problems whose clean run is correct, so the recovery curve above cannot tell us whether restoration could *rescue* failures. We therefore instrument all 100 attempted problems and record two outcomes at every restore point: whether the model emits a parseable number (output execution) and whether that number is correct (semantics). Table 5 shows the result, split by clean-run correctness.

Parseability is essentially layer-invariant: 96% even for the corrupted baseline, and 97–100% at every restore point. The model always emits a number; corruption never makes it decline. Correctness, by contrast, depends on the restored depth. For the 10 problems the clean run solves, restoration recovers 7

Table 5: Output execution versus semantic correctness. The clean residual stream is restored at one layer of the corrupted run; we record whether the model emits a parseable number (*parse*) and whether that number is correct (*acc*). Parseability is essentially layer-invariant, whereas correctness moves with the restored depth — and only for problems the model can solve without corruption. Small brackets are Wilson 95% binomial intervals; with $n = 10$ in the left group they are wide, so the recovery of that group is indicative rather than precise.

Condition	Parse rate	Accuracy [Wilson 95% CI]	
		clean correct (n=10)	clean wrong (n=90)
corrupted (no restore)	96.0%	–	1.0%
clean (no corruption)	100.0%	–	10.0%
restore layer 0	99.0%	70.0 [40, 89]%	1.1 [0, 6]%
restore layer 4	99.0%	40.0 [17, 69]%	3.3 [1, 9]%
restore layer 8	99.0%	50.0 [24, 76]%	0.0 [0, 4]%
restore layer 12	97.0%	70.0 [40, 89]%	0.0 [0, 4]%
restore layer 16	100.0%	60.0 [31, 83]%	3.3 [1, 9]%
restore layer 20	100.0%	30.0 [11, 60]%	2.2 [1, 8]%
restore layer 24	97.0%	10.0 [2, 40]%	0.0 [0, 4]%

Qwen2.5-7B-Instruct, 100 attempted problems, 7 restore points (every 4 layers of 28). Parse rate stays within 97.0–100.0% at every layer, including the corrupted baseline: the model always emits a number. Correctness, by contrast, is only recoverable for problems the clean run solves, and never recovers for the rest at any single layer.

Table 6: Same model, different answer formats. Multiple-choice recognition is easy; producing the final number is not. Task formats are not directly comparable (selection vs. generation), so the contrast is qualitative. **OlympiadBench is used only for the *direction* of the scale trend**: only a minority of its items have numerically gradable references, that subset is biased towards simpler items, and its values must not be compared numerically with SciBench.

Task	Answer format	n (gradable)	Accuracy
MMLU physics (high-school + college + conceptual)	select A–D	488	66.6 [62.3, 70.7]
SciBench physics, cot	write the number	323	21.1 [16.7, 25.7]
SciBench physics, direct	write the number	323	8.7 [5.6, 11.8]
OlympiadBench physics, cot	write the number	152 of 351	23.7 [17.1, 30.3]

Model: Qwen2.5-7B-Instruct. OlympiadBench is evaluated only on the subset whose reference answer is numerically gradable (152 of 351), which is biased towards easier items; it must not be compared numerically against SciBench.

of them (70%, Wilson 95% CI $[40, 89]$) at the best layer — suggestive but imprecise at this sample size. For the 90 problems the best layer reaches only 3 of 90 (3.3%, Wilson 95% CI $[1.1, 9.3]$), an interval whose upper bound is an order of magnitude below the correct-baseline recovery and which includes zero at several layers. The two pathways are therefore distinct, and the failures are capability limits rather than misplaced representations that a patch could repair.

What the accuracy number hides here. The accuracy metric collapses "emitted a number" and "emitted the right number" into one bit, and in this setting the first is nearly always true while the second is mostly false. Measured separately, they behave differently under intervention: format compliance is already at ceiling before any restoration and stays there, whereas correctness moves with depth. A single number cannot express that dissociation — and an evaluation that silently discards unparseable outputs would understate how rarely the parseability channel actually fails.

4.4 Knowledge or execution? A format control

Table 6 holds the model and the subject matter fixed and varies only the answer format.

The same Qwen2.5-7B gets 66.6% of 488 multiple-choice physics items right but only 21.1% when it must produce the number itself. Since the multiple-choice items test the same physics, missing knowledge cannot be the explanation: the knowledge is retrievable, the numerical execution is not.

Table 7: Error-type distribution of the SciBench cot runs, annotated independently by a *same-family* judge (Qwen2.5-14B) and a *cross-family* judge (Llama-3.1-8B); the two blocks are separated by the midrule. *unscorable_ref* (reference answer not numerically gradable) and *unparsed_pred* (no parseable answer) samples are excluded. E1 = wrong physical law; E2 = algebraic / arithmetic / unit error; E3 = violated constraint / internal inconsistency; N = grader false negative.

Model	Judge	Errors	E1	E2	E3	N
Qwen2.5-7B	Qwen2.5-14B (same family)	255	29.4%	57.6%	10.2%	2.7%
Qwen2.5-14B	Qwen2.5-14B (same family)	240	22.5%	65.0%	8.3%	4.2%
Qwen2.5-7B	Llama-3.1-8B (cross family)	255	9.0%	82.0%	9.0%	0.0%
Qwen2.5-14B	Llama-3.1-8B (cross family)	240	7.5%	85.4%	7.1%	0.0%

The 7B/14B comparison is run under *both* judges: same-family $\chi^2 = 4.55$, $df = 3$, $p = 0.208$; cross-family $\chi^2 = 1.09$, $p = 0.778$ — no significant difference either way. Agreement between the two judges is only $\kappa = 0.13$ (60% raw agreement, $\kappa_w = 0.12$ linearly weighted). We therefore claim only what survives both annotators: the modal category is E2 under both judges and E3 stays small; the precise E1/E2 split is *not* claimed, since the judges differ by a factor of three on E1.

OlympiadBench is used *only* to check the direction of the scale trend: accuracy rises from 23.7% to 30.3% (+6.6 points), consistent with SciBench. Because its 43% gradable coverage is biased towards items with clean scalar answers, we make no absolute-level claim from this benchmark and never compare its values numerically against SciBench; per-sub-domain counts are so small (six gradable optics items, all incorrect) that we do not interpret them.

4.5 Validity bound I: which stage fails, and can anyone tell?

We annotated every SciBench chain-of-thought error with a three-way scheme — E1 wrong physical law, E2 algebraic/arithmetic/unit error, E3 violated constraint — and repeated the annotation with a second, cross-family judge on the same samples.

The finding is about observability, not about our judges. The reason two competent judges disagree here is mechanical: applying the wrong law usually produces wrong arithmetic as well, so the *observable text* is compatible with either label. E1 and E2 are not separated by anything a judge can see in the generation. The per-class pattern confirms it — the judges agree 88% of the time on E2 but only 16% on E1, i.e. agreement is high exactly where the category is defined by surface evidence (an arithmetic slip inside a correct derivation) and collapses where it would require inferring an unobservable intention. This is a limitation of the taxonomy as a measurement instrument, not an indictment of either model: no annotator, human or LLM, can recover a distinction the text does not carry.

What survives the disagreement is therefore what both judges can see: E2 is the modal category and E3 is rare under *both* labellings, and the 7B/14B distributions do not differ under either (same-family $\chi^2 = 4.55$, $p = 0.208$; cross-family $\chi^2 = 1.09$, $p = 0.778$). Overall agreement is $\kappa = 0.133$ (60% raw, $\kappa_w = 0.118$ linearly weighted). The larger model gets more problems right rather than failing differently, which matches the format control: the bottleneck is execution, not physical knowledge.

What the accuracy number hides here. "Error type" is often reported as a single distribution from a single judge. With two judges the distribution shifts by more than 20 points on the E1/E2 split, which means the reported shares are partly a property of the annotator. The 7B/14B *comparison*, by contrast, is stable across judges — so the same table supports one claim robustly and another not at all, and only the agreement measurement reveals which is which.

4.6 Can prompt scaffolding repair the instability?

If the paraphrase instability were caused by a missing solution scaffold, telling the model to write down the knowns and unknowns explicitly should help. At a matched token budget on the same 150 problems it changes accuracy from 24.0% to 26.0% (+2.0 points, McNemar $p = 0.700$) — no reliable gain (Table 8). The failure is a capability property, not a missing scaffold.

Table 8: Prompt scaffolding as a substitute for capability. The same 150 problems are evaluated with a matched budget (2048 new tokens) under two prompts: standard chain-of-thought, and a structured prompt that additionally asks the model to write down the knowns and unknowns explicitly. Accuracy does not improve significantly, so the instability documented in Table 2 is not a missing-scaffold problem.

Prompt	Accuracy	95% CI	No `ANSWER:`	Output chars (mean / max)
standard cot	24.0%	[17.3, 31.3]	71%	2377 / 4224
structured (knowns/unknowns)	26.0%	[19.3, 33.3]	90%	2373 / 4639
paired difference: +2.0 pp, McNemar $p = 0.700$				
Accuracy by output-format compliance (both prompts pooled)				
outputs with `ANSWER:` line (n=58)	20.7%	–	–	–
outputs without it (n=242)	26.0%	–	–	–

Upper block: scaffolding changes accuracy by less than one percentage point and is far from significant. Lower block: most outputs ignore the requested `ANSWER:` format and are graded through the fallback extractor; accuracy is similar with and without the marker, so the fallback does not visibly bias the headline numbers, while output lengths stay well below the token budget — a format-compliance failure rather than truncation.

4.7 Validity bound II: an output-format ceiling

The second bound is on the evaluation apparatus, and it is large enough to change conclusions. Between 71% and 90% of outputs never emit the requested `ANSWER:` line. Output lengths stay far below the token budget, so this is a format-compliance failure, not truncation — the models simply do not follow the formatting instruction.

Two grading policies are then available for the same generations: strict (count format violations as failures) and fallback (recover a number from the last line). The lower block of Table 8 pools both prompts and compares outputs with and without the marker: accuracy is similar in the two groups, so the fallback does not visibly inflate the headline numbers. That is a relief for our own results but not for the field: on identical generations the two policies give different ceilings, and which one a paper used is usually not reported. A strict-format evaluation of the same models would therefore publish a lower number without disclosing that the difference is compliance, not capability — the same failure mode as the annotation bound above, in a second place.

5 Discussion

The metric is the finding. We set out to audit what a headline accuracy number can support, and in each of the three places we probed it, the number concealed a distinction that changes the reading. *Value vs. correctness:* the paraphrase effect is 30.4% in value-agreement terms and 5 of 240 items in correctness terms — an order of magnitude apart, and the larger of the two is the misleading one. *Execution vs. semantics:* parseability sits at 96%–100% regardless of intervention while correctness tracks the restored depth, so a metric that folds the two together cannot say which channel failed. *Localisation:* the data licence a band (layers 4–14) but not a layer, and the difference between those two claims is invisible in any point estimate.

Two further bounds are properties of the apparatus rather than of the models: two LLM judges agree at only $\kappa = 0.133$ on an error taxonomy whose most-talked-about category boundary is not textually observable, and 71%–90% of outputs violate the requested answer format, so strict and fallback grading report different ceilings on the same generations. Taken together, the honest summary of this literature's standard evidence — one accuracy number per model-prompt pair — is that it is *under-specified*: several of the distinctions we care about are recoverable, but only if the evaluation is designed to expose them.

What the model evidence supports. Within those bounds, the measurements are consistent: (i) accuracy on SciBench physics tops out at 25.7% and gains only 4.6 points from a two-fold parameter increase, while chain-of-thought multiplies accuracy by 2.28–2.18× for the Qwen family, so the prompt interaction dominates the scale effect in this setting; (ii) the same model reaches 66.6% on multiple-choice MMLU physics but only 21.1% when it must produce the number itself, which rules out missing knowledge: it is present and retrievable while numerical execution is not; (iii) the paraphrase instability is model-generic (33.8%, 31.7%,

24.4% in the other three models) and is value-level rather than correctness-level; (iv) restoring clean residual states recovers behaviour over an early-middle band (4–14 of 28) that we can localise as a range but not as a single peak, and the recovery collapses after layer 18. (v) Separating output execution from correctness shows parseability is layer-invariant (96% even under corruption) while correctness is depth-dependent, and on problems the model cannot solve restoration never exceeds 3.3% at any single layer. The failures are capability limits, not misplaced representations that a layer patch could repair.

Why the error taxonomy is itself uncertain (and why that is a contribution). Our cross-annotator agreement ($\kappa = 0.133$, $\kappa_w = 0.118$ linearly weighted, 60% raw agreement over 495 samples) is low, and this is not merely a nuisance for our own analysis — it reveals a fundamental limitation of the taxonomy itself. The per-class breakdown is diagnostic: the judges agree 88% of the time on E2 but only 16% on E1. That is exactly what one expects if the boundary between "misapplying a physical law" and "algebraic miscalculation" is intrinsically ambiguous: when a model applies the wrong law, the resulting arithmetic is wrong too, so the observable text is consistent with either label. We therefore report only the coarse, annotator-robust statements — E2 is the modal category under both judges, E3 is rare, and 7B/14B do not differ under either labelling ($p = 0.208$ same-family, $p = 0.778$ cross-family) — and deliberately do not quote precise category shares. This suggests that evaluating reasoning purely through coarse error categories is insufficient: taxonomy-based analysis inherits an annotator-dependent ceiling that stays invisible unless agreement is measured. We release our dual-annotated sample (495 wrong answers from two models, each labelled independently by a same-family and a cross-family judge) as a resource not only for physics reasoning but for research on LLM-as-a-judge attribution, where the *disagreement* between judges is the signal rather than noise.

Implications for mitigation. Two findings are directly actionable. First, since the dominant failure mode is arithmetic rather than physical (E2), the highest-leverage interventions are execution-side — tool use, code interpreters, or explicit symbolic bookkeeping — rather than more physical pre-training data. Second, since 30.4% of answers move under a semantics-preserving paraphrase, single-prompt evaluation understates the variance of a model's competence; reporting paraphrase-consistency alongside accuracy is a cheap and informative robustness metric. We also tested whether the instability is recoverable at inference time by adding structural scaffolding to the prompt: at a matched token budget the change is +2.0 points (McNemar $p = 0.700$), so prompt-level scaffolding does not substitute for capability, which further argues that the mitigation has to be execution-side rather than wording-side.

Threats to validity. All numbers come from a single GPU class (A800-80G) and a `transformers` backend without continuous batching, so wall-clock times are not comparable to a vLLM deployment, although greedy decoding makes accuracies backend-independent. Causal tracing covers one open-weight model and the strict-inclusion subset (14 of 150 attempted), so the layer localisation is a qualitative early-middle pattern rather than a precise per-layer estimate; the layer-parse analysis rests on 10 correct-baseline problems, which we accompany with Wilson intervals. OlympiadBench results refer to the 43% of problems with numerically gradable references and are used only for the direction of the scale trend. Error-type shares are annotator dependent ($\kappa = 0.133$) and are reported coarsely. All models are evaluated zero-shot with a single fixed prompt per condition; prompt sensitivity is itself one of the phenomena measured, so no per-model prompt tuning is attempted.

6 Limitations

Statistics of the causal probe. The causal analysis rests on 14 strictly-included problems, and the layer-parse analysis on 10 correct-baseline problems. Both are small: we report bootstrap intervals for the recovery curve and Wilson intervals for the small proportions, we do not claim a single peak layer, and we scope the conclusions to problems the model can already solve.

Annotator dependence. Error-type shares are annotator-dependent ($\kappa = 0.133$, $\kappa_w = 0.118$) and are reported coarsely for that reason. The cross-family judge is still an LLM, so this is a check against family-specific bias, not a substitute for human annotation.

Benchmark coverage. OlympiadBench numbers refer to the 43% of problems with numerically gradable references and are used only for the direction of the scale trend.

Backend and hardware. All runs use a single A800-80G with the `transformers` backend and no continuous batching, so wall-clock times are not comparable to a vLLM deployment; greedy decoding makes accuracies backend-independent.

Probe construction. The paraphrase probe uses deterministic rule-based rewriting; a model-generated paraphrase would be stronger but introduces rewrite quality as a confound. The scaffolding probe used one prompt design at one budget, so its null result is evidence against that intervention, not proof of impossibility. Finally, all models are evaluated zero-shot with a single fixed prompt per condition, and no per-model prompt tuning is performed, because prompt sensitivity is itself one of the phenomena measured. Note that the two conditions of the scaffolding comparison are *each* a single fixed prompt (standard chain-of-thought and the structured variant) evaluated at a matched token budget, so the comparison isolates the prompt content rather than the budget.

A The *swap* probe

The *swap* perturbation exchanges two distinct physical quantities, producing a physically inconsistent problem. Most benchmark problems contain quantities of a single type or carry no units, so only a minority of items admit such a swap. Because the resulting sample is small and its percentages therefore unstable, we report the numbers for completeness and draw *no* quantitative conclusion from them; they are excluded from Table 2 precisely so that they are not read as comparable to the other two probes.

Table 9: The *swap* probe, reported for completeness only. Coverage is a property of the data (few items admit a two-quantity exchange), not of the model, and is identical across models.

Model	Coverage	n parsed	Median sens.	Changed
Qwen2.5-3B	36/323 (11.1%)	36	0.840	83.3%
Qwen2.5-7B	36/323 (11.1%)	36	0.371	69.4%
Qwen2.5-14B	36/323 (11.1%)	36	0.533	80.6%
Llama-3.1-8B	36/323 (11.1%)	36	0.335	63.9%

B Prompt templates

All prompts are rendered from a single template table in `src/prompts.py`; the MCQ templates are selected automatically when the question contains at least two options of the form "A. . . .". Every template demands the final answer on its own line prefixed with `ANSWER:`, which is what the extractor looks for first.

Cot / free-response

```
Solve the following physics problem step by step. Show all work.
At the end, give the final numeric answer on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Cot / multiple-choice

```
Solve the following multiple-choice physics problem step by step. Show all work.
At the end, give the chosen option letter on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Direct / free-response

```
Answer the following physics problem directly with a single number (no explanation).
Give the final answer on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Direct / multiple-choice

```
Answer the following multiple-choice physics problem.
Give the chosen option letter on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Structured / free-response

```
Solve the following physics problem step by step.
Before solving, first write down the knowns and unknowns explicitly, then derive the solution.
Show all work. At the end, give the final numeric answer on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Structured / multiple-choice

```
Solve the following multiple-choice physics problem step by step.
Before solving, first write down the knowns and unknowns explicitly, then compare the options.
At the end, give the chosen option letter on its own line, prefixed with "ANSWER:".
Problem: <problem statement>
```

Chat template The rendered prompt is then wrapped by the tokenizer's chat template (`apply_chat_template`), so the model sees its own instruction format. Whether a chat template was applied is recorded in every result file as `meta.chat_template`. Example problem used in this paper: `A 2.0 kg block slides down a 30 degree incline at 3.0 m/s. ...`

C Result-file schema

Every evaluation writes one JSON file. The top level records the run identity and provenance; `detail` holds one record per problem, which is what the audit tables are built from. Minimal schema of `results/eval_*.json`:

```
  "model": string,
  "benchmark": string,
  "mode": string,
  "n": int,
  "n_scorable": int,
  "n_unscored": int,
  "n_unparsed_pred": int,
  "overall_acc": string,
  "per_domain_acc": object,
  "detail": [ ... 323 records ... ],
  "wall_time_s": float,
  "max_new_tokens": int,
  "data_composition": object,
  "prompt_fingerprint": string,
  "meta": object,
  # detail[i]:
  #   "id": string,
  #   "question": string,
  #   "domain": string,
  #   "subset": string,
  #   "ref": string,
  #   "pred_raw": string,
  #   "pred_val": float,
  #   "correct": bool,
  #   "status": string,
  #   "scorable": bool,
  #   "n_pred_chars": int,
```

Two fields deserve emphasis. `scorable` marks whether the *reference* answer could be parsed; rows with `scorable=false` are excluded from every denominator and reported as `n_unscored`. `status` is one of `correct`, `incorrect`, `unparsed_pred` or `unscorable_ref`, so a reader can recompute any accuracy in this paper from the raw file without re-running a model. Sensitivity files (`sensitivity_*_*.json`) additionally carry `items[]`, one entry per perturbed problem with the perturbed question text and both answers, enabling per-item audit of the perturbation.

D Statistical details

Bootstrap. Confidence intervals resample problems with replacement $B = 10{,}000$ times using `random.Random` seeded with 42, and take the 2.5/97.5 percentiles of the resampled statistic (order statistics of the resampled accuracy, i.e. a percentile interval), the standard non-parametric approach of Efron and Tibshirani [1]. The same seed is used for every model, so intervals are exactly reproducible across machines and do not depend on the NumPy version.

Layer-wise interval and paired test. The recovery curve has $n = 14$ problems. Each layer's mean gets a percentile bootstrap interval from the same B; to test whether the point estimate's best layer is meaningful, we bootstrap the *difference* between that layer and every other layer (2,000 resamples) and ask whether the interval excludes zero. With this test the best layer (layer 8) is *not* separable from 6 other layers, which is why the paper claims a band rather than a peak.

Small-sample proportions. Proportions computed on few observations are reported with Wilson score intervals (95%), which behave better than the normal approximation near 0 and 1. For example the layer-parse recovery on the correct-baseline group is a proportion of a 10-problem sample.

Paired tests. Correctness comparisons between two conditions on the same problems use McNemar's test [7] on the discordant pairs. When the number of discordant pairs is small we report the exact binomial two-sided p-value rather than the asymptotic χ^2 version.

Annotator agreement. We report both unweighted Cohen's κ and a linearly weighted κ_w [6] that gives partial credit for adjacent categories, plus per-class agreement. The distinction matters because the categories are heavily imbalanced (E2 dominates), which depresses κ even when most individual decisions agree.

Table 10: Per-class agreement between the same-family and cross-family judge ($n = 495$ samples; counts are per class).

Class	cross-family count	same-family count	both	agreement given same-family label
E1	129	41	20	16%
E2	303	414	267	88%
E3	46	40	11	24%
N	17	0	0	0%

E Reproducibility

Code and version. All code, evaluation scripts, and analysis utilities are released in the accompanying repository (`https://github.com/380992846-tech/EMPR`) at the exact commit used for this paper (`ec5d0c02c35660a3ac83fc27a1b67de9f0b9df6d`). Defaults live in a single configuration file (`config.yaml`); each analysis stage has a single entry point. Every result file records its own provenance in a `meta` block (`git_commit`, execution environment, `prompt_fingerprint`, `max_new_tokens`, `chat_template`, `wall_time_s`), so any number in the paper can be traced to the run that produced it.

Datasets. **SciBench** physics (`https://github.com/mandyyyyii/scibench`; subjects matter, class, fund, diff, quan, thermo) yields $N_{sci} = 323$ problems, of which mechanics 56, thermodynamics 66, optics 47, math 50, quantum 33, general 71. All 323 reference answers are numerically gradable, so no sample is excluded

from the denominator of the main table. **OlympiadBench** physics (`https://huggingface.co/datasets/Hothan/OlympiadBench`; configs `OE_TO_physics_en_COMP` and `OE_TO_physics_zh_CEE`) yields 351 problems, of which only 152 (43.3%) have a numerically gradable reference answer; the remaining 199 are symbolic or multivariate solutions and are excluded from the denominator (reported as `n_unscored`). **MMLU** physics (`cais/mmlu`; high-school, college and conceptual physics) yields 488 multiple-choice problems. All datasets are public and are fetched by `scripts/download_data.py`; SciBench is loaded from the local `data/` copy when present so that the exact same 323 problems are used on every machine (`EMPR_PREFER_HF=1` overrides this).

Models. All models are open-weight and were run from public checkpoints: Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Meta-Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. No closed-weight or API-only model contributes to any reported number. Causal tracing is performed exclusively on an open-weight model whose residual stream is accessible (Qwen2.5-7B-Instruct). Models listed in `config.yaml` but not reported here were not run and no claim is made about them.

Inference settings. Decoding is **greedy** (temperature = 0) with a fixed seed (42), so all outputs are deterministic. Maximum new tokens were 1024 for chain-of-thought prompts (1536 for the reasoning-distilled model, whose chains are longer) and 96 for answer-only prompts (64 for multiple-choice and for the sensitivity probes). Inference used the `transformers` backend on a single GPU in bfloat16 with batches of 16–48; vLLM was not available in the execution environment, so the transformers path was used throughout and no result mixes backends. The tokenizer's chat template is applied to every instruct model (recorded as `meta.chat_template`) and batch generation uses left padding.

Answer extraction and grading. Correctness is judged by symbolic/numeric comparison. The final answer is extracted by a precedence-ordered procedure (`ANSWER:` line, taking the *last* occurrence; then `\boxed{}`; then "answer is X"; then the last number on the last line), normalised for Unicode minus signs, superscripts, `\times` and thousands separators, and compared to the reference under a relative tolerance of 5% (absolute tolerance 0.01 for near-zero targets) via the Python/SymPy evaluator. Reference answers are parsed with a *stricter* parser than model outputs: a reference that is not a single number (up to units) is marked `unscorable_ref` and removed from the denominator rather than counted as a model error, because a permissive parser would read `x^2+2x` as 2 and silently corrupt the denominator. Grading is covered by 23 self-test cases (`python -m src.grading`, 23/23 pass) and by the unit-test suite (84 tests). As an end-to-end check, using the reference answers as model outputs yields 323/323 self-consistent judgements. No LLM judge is used for the primary results.

Statistical protocol. All reported accuracies carry bootstrap 95% confidence intervals from 10,000 resamples with seed = 42. The bootstrap is implemented in pure Python using `random.Random` and order statistics, so the intervals are reproduced *identically* across machines, independent of the NumPy version. Pairwise comparisons use McNemar's test (on paired per-problem outcomes); error-distribution comparisons use an uncorrected Pearson chi-square; agreement between the two error annotators is quantified with Cohen's κ.

Sensitivity probes. Three deterministic perturbations are applied to the same 323 SciBench problems: *numeric* (scale physical values; physics intact; 305/323 problems affected), *wording* (surface paraphrase only, physics and data bit-identical; 240/323), and *swap* (exchange two distinct physical quantities, creating a physically inconsistent problem; 36/323). Because the correct answer to a *numeric* or *swap* variant is not directly computable, only the distribution of output change is reported for those variants; *wording* keeps the reference answer valid, so it additionally supports a paired accuracy comparison. Median and p90 sensitivity are reported instead of the mean, since the score $|\Delta|/(|base| + \epsilon)$ is dominated by outliers when $|base| \approx 0$.

Causal tracing. We follow the corrupt-and-restore protocol: the numeric quantities of a question are replaced by a semantically empty placeholder (C2 corruption), the corrupted prompt is run, and the clean residual stream is restored at exactly one layer at a time; all other layers remain corrupted, isolating each layer's contribution. Restoration is applied only during prefill (never to single-token decode steps), and the

Table 11: Compute breakdown for all reported runs (single A800-80G, transformers backend, bf16, greedy decoding). Times are GPU-busy time; model download and orchestration are excluded.

Stage	Runs / scale	GPU time
SciBench, chain-of-thought	5 models × 323	1.05 h
SciBench, answer-only	4 models × 323	0.02 h
OlympiadBench, chain-of-thought	2 models × 351 (152 gradable)	0.58 h
MMLU physics, multiple choice	1 model × 488	0.002 h
Scaffolding comparison	1 model × 150 × 2 prompts (2048 tok)	0.57 h
Sensitivity probes	4 models × 323 × 3 variants	0.10 h
Causal tracing, valid-only	1 model, 150 attempted (14 valid)	0.05 h
Causal tracing, all-attempts	1 model, 100 attempted × 7 layers	0.08 h
Error annotation	2 judges × 2 models × 495 samples	0.17 h
Total	25 runs	**≈2.6 h**

clean and corrupted sequences are aligned from the right because they tokenise to different lengths. Only problems whose clean run was *correct* and whose corrupted run *lost score* enter the recovery rate; of 150 attempted problems, 14 qualify (134 skipped as clean-wrong, 2 as corruption-flat), which is the sample size behind the reported curve.

Hardware and compute. All runs were executed on a single NVIDIA A800-80G PCIe GPU (112 vCPU, 991 GB RAM, 50 GB scratch disk) with bf16 weights. Checkpoints were downloaded one at a time and deleted after use, so peak scratch usage stayed below 30 GB; a single 14B checkpoint needs ≈30 GB free. Total compute for all reported runs is **≈2.6 GPU-hours**, of which ≈2.2 h is the fourteen evaluation runs (summed from the `wall_time_s` recorded in each result file) and the remainder is sensitivity probes, causal tracing, the scaffolding comparison and error annotation. Table 11 gives the breakdown. An additional ≈1.0 GPU-hour was spent on one full-scale causal-tracing run that was aborted (it was too slow to be practical at full layer resolution) and then re-run at reduced scale; it contributes no reported number.

Reproducibility checklist.

- Deterministic decoding (temperature 0, seed 42): ✓
- Public datasets + scripted download, pinned local copy: ✓
- Open-weight models, public checkpoints: ✓
- Symbolic/numeric grading with 23-case self-test and 83 unit tests: ✓
- Reference-answer strictness separated from output-extraction leniency: ✓
- Bootstrap 95% CI, machine-independent (pure Python): ✓
- Causal tracing on an open-weight model only: ✓
- Code, config and per-run metadata at a pinned commit: ✓
- Error-type annotation cross-checked with a second, cross-family judge: ✓

Limitations affecting reproducibility and interpretation. (i) The `swap` perturbation covers only 36/323 problems (11.1%) because many problems contain quantities of a single type or carry no unit; this is reported rather than artificially inflated, and the swap results are treated as supporting evidence only. (ii) OlympiadBench is evaluated only on the 43.3% of problems whose reference answer is numerically gradable; that subset is biased towards easier items, so OlympiadBench accuracies must not be compared numerically with SciBench accuracies. (iii) Error-type shares are annotator-dependent: a same-family judge (Qwen2.5-14B) and a cross-family judge (Llama-3.1-8B) agree on only $\kappa = 0.133$ (60.2% raw agreement) on the same

495 samples. Both judges identify E2 (algebraic/computational error) as the dominant category and E3 as rare, which is what we claim; the precise E1/E2 split is not claimed, because the E1/E2 boundary is intrinsically fuzzy (applying the wrong law usually also shows up as an arithmetic error). (iv) Unparsed model outputs are counted and reported (`n_unparsed_pred`), never imputed; on SciBench this count is zero for all reported runs. A related and more consequential limitation is *format compliance*: between 71% and 90% of chain-of-thought outputs ignore the explicitly requested `ANSWER:` line and are graded through the fallback extractor ("last number on the last line"). Output lengths stay well below the token budget (mean ≈2,400 characters, max ≈4,600 against a 2048-token cap), so this is not truncation, and accuracy does not differ systematically between compliant and non-compliant outputs (0.21 vs. 0.25 in the matched subset), i.e. the fallback does not obviously bias the headline numbers. It nevertheless means that any evaluation relying on strict output formatting would measure a lower ceiling than ours. (v) Throughput figures refer to the transformers backend; vLLM would be 5–10× faster but is not used here, so wall-clock times are not comparable across backends. (vi) Prompt-level scaffolding does not substitute for capability: adding an explicit "write down the knowns and unknowns" instruction changes matched-budget accuracy by +2.0 percentage points (McNemar $p = 0.70$, same 150 problems, 2048-token budget), so we do not claim that prompt engineering repairs the instability documented by the paraphrase probes.

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[TODO: funding / compute providers / reviewers]

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